

Mahsa Ebrahim

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EDUCATION

PhD. Mechanical Engineering, Villanova university (Aug. 2011- Present)

Thesis title: Identification of the Impact Regimes of a Liquid Droplet Propelled by a Gas Stream Impinging onto a Heated Surface

Advisor: Dr. Alfonso Ortega

GPA: 4.0/4.0

Visiting Researcher, University of Leeds (Fall 2015)

Institute of ThermoFluids: Multi-Phase lattice Boltzmann modeling

Advisor: Dr. Jonathan Summers

M.Sc. Mechanical Engineering, K.N.T.U University (www.kntu.ac.ir)

(Sep.2006 - Sep.2009)

Thesis title: Development of Fast and Robust Grid Generation Techniques for CFD applications

Advisor: Dr. Ali Ashrafizadeh

GPA: 17.3/20

B.Sc. Mechanical Engineering, K.N.T.U University (www.kntu.ac.ir)

(Sep.2002- Sep.2006)

Thesis title: Micro Fluidics in Thermodynamic and Heat Transfer Point of View

Advisor: Dr. Majid Ghasemi

GPA: 17.8/20

HONORS

- 1- Ranked **1st** among 30 students in mechanical engineering department, M.Sc., K.N.T.U University
- 2- **Invited to study Master of Science in K.N.Toosi University having the Iranian nation-wide entrance exam waved** (20000 candidates)
- 3- Ranked **1st** among 120 students in mechanical engineering department, B.Sc., K.N.T.U University
- 4- Ranked 2244 out of 310000 candidates of nation-wide entrance exam for Bachelor of Science in Iran

PUBLICATIONS

Journals

- 1- **Ebrahim, M.** and Ortega, A., "Identification of the Impact Regimes of a Liquid Droplet Propelled by a Gas Stream Impinging onto a Dry Surface," Journal of Applied Physics, 2014 (Under preparation)

Conferences

- 1- **Ebrahim, M.**, Silva, L., and Ortega, A., "Heat Transfer due to the Impingement of an Axisymmetric Synthetic Jet Emanating from a Circular Orifice: A Numerical Investigation of a Canonical Geometry," Proc. of the 2013 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems (InterPACK2013)
- 2- Silva, L., **Ebrahim, M.**, and Ortega, A., "Experiment and Simulation of Impinging Synthetic Jet Designed to Eliminate Actuator Artifacts," Proc. of the ASME 2012 Summer Heat Transfer Conference HT2012
- 3- **Ebrahim, M.**, Ashrafizadeh, A. and Ghasemi, M., "Development of an Algebraic Grid Generation for CFD applications", ISME2009
- 4- Ashrafizadeh, A., **Ebrahim, M.**, Jalalabadi, R., Structured Grid Generation (SGG) Workshop, University of Waterloo, 03/27/2009

Book Chapter

- 1- Ashrafizadeh, A., **Ebrahim, M.** and Jalalabadi, R. (2012). Alternative Methods for Generating Elliptic Grids in Finite Volume Applications, Finite Volume Method - Powerful Means of Engineering Design, PhD. Radostina Petrova (Ed.), ISBN: 978-953-51-0445-2, InTech

EDITORIAL ACTIVITIES

Reviewer for ASME conferences

- 1- InterPACK 2013
- 2- IEEE-ITHERM 2014
- 3- IMECE 2014

RESEARCH INTERESTS

Convection heat transfer, Fluid mechanics, Computational fluid dynamics, Thermodynamic modeling

TEACHING EXPERIENCE

Teaching Assistant

Thermal- Fluid Design (ME4850)- Villanova University (Spring 2014)

- Teaching 3 sessions of Thermal radiation While being observed by faculty member and VITAL (Villanova Institute for Teaching and Learning)

- Holding office hours for ongoing projects of class

Fluid Mechanics- K.N.Toosi University (2006-2007)

- 3 hours of problem solving session per week
- Teaching 4 sessions
- Being responsible for homework, exams and grading

Heat Transfer- K.N.Toosi University (2006-2007)

- 3 hours of problem solving session per week
- Teaching 6 sessions
- Being responsible for homework, exams and grading

Instructor

Thermal Fluids lab (ME4003)- Villanova University (Fall 2013)

- Being responsible for Jet Engine and Steam Turbine

Certificate

Certificate of **Teaching Engineering in higher Education**- Villanova University (Spring 2014)

WORKING EXPERIENCE

SAZEH Consultants- Piping and Stress analysis and Equipment design department

Senior Engineer (2010-2011) – **Promoted after 2 years**

- **South Pars**- phase 12 (Deputy of Project Senior Leader)
 - Basic and detail design leader of piping of Slug catcher unit
 - Stress analysis and supporting of piping
 - Data sheet preparation of Pumps and Tanks
 - Design leader of firefighting and safety lines
 - Design leader of underground cooling water supplies, safety and drain lines
 - Responsible for the review meetings with the Project Clients
 - Final checking of outgoing documents
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- **Arak Refinery** plant- Unit 18 (Construction Project Senior Leader)
 - Site visit 2 days per week- Fall 2010
 - Leader Engineer for design changes regarding to construction problems

Junior Engineer (2008-2010)

- **Arak Refinery** plant- Unit 18
 - Detail design of piping system

- Pipe Rack designer
 - Stress analysis and supporting of piping zone around a compressor
 - Data sheet preparation of Pumps and Vessels and Towers
 - Firefighting and safety lines designer
 - Design leader of underground cooling water supplies, safety and drain lines
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- **BBII Gas plant- Units 02 and 18**
 - Detail design of piping system
 - Pipe Rack designer
 - Stress analysis and supporting of piping
 - Data sheet preparation of Tanks and Cooling Tower
 - Firefighting and safety lines designer
 - Design leader of underground cooling water supplies, safety and drain lines

COMPUTER SKILLS

- Software: FLUENT, Mathematica, Mathcad, SolidWorks, AutoCad, LabView, Tecplot, PDMS
- Programming: MATLAB, FORTRAN

LANGUAGE SKILLS

- English- Fluent in speaking, listening and writing
- Farsi- Native language